

Azure FinOps Best Practices

Validation & Remediation Report

Scope: Contoso — Sample Production Subscription

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Rule catalog version: 1.0.0

Resources evaluated: 11 | Findings: 21

Aligned to the FinOps Foundation FinOps Framework, the Azure Well-Architected Framework Cost Optimization pillar, and Azure Advisor cost recommendation categories.

Executive Summary

Metric	Value
Resources evaluated	11
Rules evaluated against inventory	17
Rules requiring additional account-scope data	5
Total findings	21
High severity findings	5
Medium severity findings	12
Low severity findings	4
Monthly cost on flagged resources	\$10,178.80

Note: 5 rule(s) were not evaluated because the submitted inventory did not include the required account/subscription-scope signals (e.g., budgets, Advisor feed, load-balancer telemetry): AZFO-008, AZFO-017, AZFO-018, AZFO-019, AZFO-024. Recommend a follow-up pass once that data is available — see references/finops_rules.json for detection logic.

Findings Overview

Rule ID	Severity	Resource	Type	Finding
AZFO-001	High	vm-web01	Compute/virtualMachines	14-day avg CPU 2.8%, max CPU 9.5% — right-sizing candidate.
AZFO-002	High	vm-test-legacy	Compute/virtualMachines	30-day avg CPU 0.4%, minimal network activity while running — idle VM.
AZFO-003	High	vm-web01	Compute/virtualMachines	30-day uptime ratio 98% on pay-as-you-go pricing with no commitment discount.
AZFO-003	High	vmss-api	Compute/virtualMachineScaleSets	30-day uptime ratio 99% on pay-as-you-go pricing with no commitment discount.
AZFO-015	High	salesdb	Sql/servers/databases	14-day avg provisioned-capacity utilization 14%.
AZFO-004	Medium	vm-web01	Compute/virtualMachines	Eligible Windows workload without Azure Hybrid Benefit applied (licenseType=None).

Rule ID	Severity	Resource	Type	Finding
AZFO-004	Medium	salesdb	Sql/servers/databases	Eligible SQLServer workload without Azure Hybrid Benefit applied (licenseType=LicenseIncluded).
AZFO-005	Medium	disk-orphan-01	Compute/disks	Disk unattached for 45 days.
AZFO-006	Medium	disk-oversized-02	Compute/disks	Premium_LRS disk averaging 90 IOPS against 5000 provisioned.
AZFO-009	Medium	stcoldata001	Storage/storageAccounts	No lifecycle policy; 4500000000000 bytes in hot tier untouched for 30+ days.
AZFO-012	Medium	vm-test-legacy	Compute/virtualMachines	Environment tag 'dev' with no auto-shutdown schedule configured.
AZFO-013	Medium	asp-web-prod	Web/serverfarms	14-day avg CPU 8.5% across 4 static instances, no autoscale.
AZFO-016	Medium	vm-test-legacy	Compute/virtualMachines	Missing required tag(s): CostCenter, Owner, Application.
AZFO-016	Medium	disk-orphan-01	Compute/disks	Missing required tag(s): Owner, Application.
AZFO-016	Medium	pip-unused-01	Network/publicIPAddresses	Missing required tag(s): CostCenter, Owner, Environment, Application.
AZFO-016	Medium	bastion-hub	Network/bastionHosts	Missing required tag(s): CostCenter, Owner, Application.
AZFO-025	Medium	vmss-api	Compute/virtualMachineScaleSets	No autoscale profile with CPU std-dev of 27 over 14 days (variable load).
AZFO-007	Low	pip-unused-01	Network/publicIPAddresses	Standard public IP with no associated ipConfiguration.
AZFO-010	Low	stcoldata001	Storage/storageAccounts	Standard_GRS replication with DataCriticality tag = 'low'.
AZFO-022	Low	law-central	OperationalInsights/workspaces	Sustained 135 GB/day ingestion on Pay-As-You-Go pricing tier.
AZFO-023	Low	bastion-hub	Network/bastionHosts	Zero Bastion sessions in the last 30 days.

Detailed Remediation Guidance

AZFO-001 | Underutilized (right-sizing candidate) Virtual Machine

Severity	High
Category / Domain	Compute / Optimize
Affected resources	vm-web01
Remediation	Right-size to a smaller SKU in the same VM family using Azure Advisor / Azure Monitor metrics, or move the workload to Azure Container Apps / Functions if usage is bursty. Validate with a 30-day metrics baseline before resizing production workloads.
Estimated savings	20-50% of the VM's compute cost

AZFO-002 | Idle Virtual Machine (candidate for deallocation)

Severity	High
Category / Domain	Compute / Optimize
Affected resources	vm-test-legacy
Remediation	Confirm with the resource owner, then stop/deallocate or delete the VM. If retained for DR/testing, schedule automated start/stop via Azure Automation or the Start/Stop VMs v2 solution.
Estimated savings	Up to 100% of the VM's compute cost while stopped/deallocated

AZFO-003 | Missing Reserved Instance / Savings Plan coverage on steady-state compute

Severity	High
Category / Domain	Commitment Discounts / Optimize
Affected resources	vm-web01; vmss-api
Remediation	Purchase a 1- or 3-year Azure Reservation for the specific VM size/region, or a broader Azure Savings Plan for Compute for flexible coverage across VM families. Use Azure Advisor reservation recommendations to size the commitment against actual usage.
Estimated savings	Up to 65-72% versus pay-as-you-go compute rates

AZFO-004 | Azure Hybrid Benefit not applied to eligible Windows Server / SQL Server workload

Severity	Medium
Category / Domain	Licensing / Optimize
Affected resources	vm-web01; salesdb
Remediation	Enable Azure Hybrid Benefit on the VM/SQL resource (licenseType=Windows_Server or Windows_Server_Perpetual) to reuse existing on-prem licenses instead of paying the embedded license cost in the Azure meter.
Estimated savings	Up to 40% on Windows VM cost; up to 55% on SQL Server cost

AZFO-005 | Unattached Managed Disk

Severity	Medium
Category / Domain	Storage / Optimize
Affected resources	disk-orphan-01
Remediation	Confirm the disk is not held for a pending restore/migration, then snapshot if retention is required and delete the disk, or downgrade to Standard HDD if it must be retained but is rarely accessed.
Estimated savings	100% of the disk's monthly storage cost if deleted

AZFO-006 | Premium/high-performance disk tier on low-IOPS workload

Severity	Medium
Category / Domain	Storage / Optimize
Affected resources	disk-oversized-02
Remediation	Right-size to Standard SSD or Standard HDD, or move to a smaller Premium SSD provisioned-size tier that still meets the observed IOPS/throughput ceiling.
Estimated savings	40-80% of the disk's monthly cost

AZFO-007 | Orphaned / unassociated Public IP address

Severity	Low
Category / Domain	Networking / Optimize
Affected resources	pip-unused-01
Remediation	Release or delete the public IP if it is no longer required, or re-associate it with the intended resource.
Estimated savings	100% of the public IP hourly charge

AZFO-009 | Storage account without lifecycle management policy

Severity	Medium
Category / Domain	Storage / Optimize
Affected resources	stcoldata001
Remediation	Configure a Lifecycle Management policy to transition blobs to Cool/Cold/Archive tiers based on last-access time, and set expiration rules for temporary or log data.
Estimated savings	up to 50% (Cool), up to 80% (Archive) on tiered data

AZFO-010 | Redundant or excessive storage replication tier

Severity	Low
Category / Domain	Storage / Optimize
Affected resources	stcoldata001
Remediation	Downgrade to LRS or ZRS where cross-region redundancy is not a business requirement; confirm against data-residency and DR policies first.
Estimated savings	~50% of replicated storage cost

AZFO-012 | Non-production workload running outside business hours without auto-shutdown

Severity	Medium
Category / Domain	Compute / Optimize
Affected resources	vm-test-legacy
Remediation	Enable VM auto-shutdown or the Start/Stop VMs v2 automation runbook to power resources down outside business hours; consider Dev/Test pricing subscription offers for eligible workloads.
Estimated savings	~65% (assuming 12x5 vs 24x7 runtime)

AZFO-013 | App Service Plan over-provisioned relative to app load

Severity	Medium
Category / Domain	PaaS Compute / Optimize
Affected resources	asp-web-prod
Remediation	Right-size the App Service Plan tier, consolidate low-traffic apps onto a shared plan, and enable autoscale rules instead of a static high instance count.
Estimated savings	20-40% of App Service Plan cost

AZFO-015 | Azure SQL / Cosmos DB provisioned throughput far exceeds consumption

Severity	High
Category / Domain	Database / Optimize
Affected resources	salesdb
Remediation	Move to serverless tier (Azure SQL Database serverless, Cosmos DB serverless/autoscale) or right-size the provisioned tier; enable Cosmos DB autoscale to eliminate manual RU/s guesswork.
Estimated savings	30-60% of database compute cost

AZFO-016 | Missing cost allocation tags (governance / showback-chargeback gap)

Severity	Medium
Category / Domain	Governance / Inform
Affected resources	vm-test-legacy; disk-orphan-01; pip-unused-01; bastion-hub
Remediation	Apply required tags directly or via Azure Policy 'Require a tag on resources' with remediation tasks; consider Azure Policy tag inheritance from resource group/subscription to reduce manual tagging burden.
Estimated savings	Not directly cost-reducing; enables accurate allocation and downstream optimization

AZFO-022 | Log Analytics workspace using Pay-As-You-Go tier without commitment tier evaluation

Severity	Low
Category / Domain	Observability / Optimize
Affected resources	law-central
Remediation	Evaluate and switch to the appropriate Log Analytics Commitment Tier that matches sustained ingestion volume; also review verbose diagnostic categories and reduce retention on low-value tables.
Estimated savings	Up to 30% of ingestion cost at qualifying volumes

AZFO-023 | Unused or excess Azure Bastion / VPN Gateway standing infrastructure

Severity	Low
Category / Domain	Networking / Optimize
Affected resources	bastion-hub
Remediation	Deploy Bastion using the Developer SKU for occasional access, or deallocate/delete and re-provision on demand via IaC when access is actually needed.
Estimated savings	Up to 100% of the Bastion hourly charge

AZFO-025 | Autoscale not enabled on Virtual Machine Scale Set with variable load

Severity	Medium
Category / Domain	Compute / Optimize
Affected resources	vmss-api
Remediation	Configure autoscale rules based on CPU, memory, or queue-length metrics with appropriate min/max/default instance counts and cooldown periods; consider mixing in Spot instances for stateless, interruptible capacity.
Estimated savings	20-45% of scale set compute cost

Appendix: Full Rule Catalog

Complete list of rules in this validation pass, including rules not triggered by any finding. Full detection logic lives in [references/finops_rules.json](#).

Rule ID	Title	Category	Severity
AZFO-001	Underutilized (right-sizing candidate) Virtual Machine	Compute	High
AZFO-002	Idle Virtual Machine (candidate for deallocation)	Compute	High
AZFO-003	Missing Reserved Instance / Savings Plan coverage on steady-state compute	Commitment Discounts	High
AZFO-004	Azure Hybrid Benefit not applied to eligible Windows Server / SQL Server workload	Licensing	Medium
AZFO-005	Unattached Managed Disk	Storage	Medium
AZFO-006	Premium/high-performance disk tier on low-IOPS workload	Storage	Medium
AZFO-007	Orphaned / unassociated Public IP address	Networking	Low
AZFO-008	Idle / underutilized Load Balancer or Application Gateway	Networking	Low
AZFO-009	Storage account without lifecycle management policy	Storage	Medium
AZFO-010	Redundant or excessive storage replication tier	Storage	Low
AZFO-011	Stale disk/VM snapshots retained indefinitely	Storage	Low
AZFO-012	Non-production workload running outside business hours without auto-shutdown	Compute	Medium
AZFO-013	App Service Plan over-provisioned relative to app load	PaaS Compute	Medium
AZFO-014	AKS cluster node pool without autoscaling enabled	Containers	Medium
AZFO-015	Azure SQL / Cosmos DB provisioned throughput far exceeds consumption	Database	High
AZFO-016	Missing cost allocation tags (governance / showback-chargeback gap)	Governance	Medium
AZFO-017	No budget or cost alert configured at subscription/resource-group scope	Governance	Medium
AZFO-018	Azure Advisor cost recommendations outstanding and unactioned	Governance	Medium
AZFO-019	Development/Test subscription not using Dev/Test pricing offer	Licensing	Low
AZFO-020	ExpressRoute circuit or VPN Gateway provisioned above required bandwidth tier	Networking	Low

Rule ID	Title	Category	Severity
AZFO-021	Duplicate or unused Azure Container Registry / Log Analytics workspace	Governance	Low
AZFO-022	Log Analytics workspace using Pay-As-You-Go tier without commitment tier evaluation	Observability	Low
AZFO-023	Unused or excess Azure Bastion / VPN Gateway standing infrastructure	Networking	Low
AZFO-024	Multiple overlapping FinOps/cost tools or exports without a single source of truth	Governance	Low
AZFO-025	Autoscale not enabled on Virtual Machine Scale Set with variable load	Compute	Medium